Title 0:00–0:15. Joint work with Ken Brown.

CSS codes and transversal gates

0:15–1:00 / 1 Outline Preview the $[\![8,3,2]\!]$ example, conditions, geometry, completion, and bounds.

1:00–2:00 / 2 Generator rows Columns labelled by cube points. Rows u_1 = 00001111, u_2. Overlap 00000011 is an edge. XOR.

2:00–3:00 / 3 Pauli strings $X(u_1)$ on a face. Overlap 2 commutes, overlap 1 anticommutes.

3:00–4:15 / 4 Stabilizer, CSS Syndrome, $[\![n,k,d]\!]$, $d = \min(d_X, d_Z)$. $X(a)$ for $a \in C_2$, $Z(b)$ for $b \in C_1^{\perp}$. $G = [K;S]$, $|x\rangle_L$.

4:15–5:45 / 5 $[\![8,3,2]\!]$ code Rows $K_i = u_i$, $S = \mathbf{1}$. Pairs 00001111, 11110000, $x = 111$. Z on 000: syndrome 1. Edge: logical, d_Z=2, d_X=4.

5:45–6:45 / 6 Clifford, non-Clifford Gates on basis states. CCZ flips $|111\rangle$. Add T or CCZ for universality.

6:45–7:30 / 7 Transversal, distillation One gate per qubit. Eastin–Knill: not universal. Distillation needs transversal T or CCZ.

7:30–8:35 / 8 Three codes Steane: H, S. 15-qubit: $T^{\otimes 15} = \bar{T}^{\dagger}$. $[\![8,3,2]\!]$: T/T† gives CCZ. State the question.

$[\![8,3,2]\!]$ example *8:35–8:45: outline, current theme highlighted.*

8:45–9:45 / 9 Signed weights Integer signed weight, mod 8. Every codeword in a logical coset needs the same phase.

9:45–11:45 / 10 Pattern from overlaps Ask where $T^{\dagger}$ goes. Reveal $\gamma = u_1 + u_2 + u_3$. Face 0, edge 0, vertex -1.

11:45–12:30 / 11 Row congruences Coefficients 1, -2, 4 give mod 8, 4, 2. Only the logical triple is odd.

Gate conditions *12:30–12:40: outline, current theme highlighted.*

12:40–13:40 / 12 Triorthogonal codes Timeline. Triorthogonal: even overlaps, but CCZ needs one odd triple.

13:40–15:10 / 13 Nine CH conditions Logical, mixed, stabilizer families. K3 odd. Parities allow a Clifford correction.

Geometric construction *15:10–15:20: outline, current theme highlighted.*

15:20–16:40 / 14 Q48 result 48 physical, 3 logical. d_X=16, d_Z=3. 26 T and 22 inverse, no correction.

16:40–17:40 / 15 Fix the stabilizers Projective columns. 8-divisible span. Find K. Solve signs. Divisibility is a choice.

17:40–18:40 / 16 Projective syndromes Nonzero columns detect one Z. Distinct columns detect two. This proves only d_Z.

18:40–19:40 / 17 48-point support 63 minus 15 gives 48. Remove w = 00. Three blocks of 16. Six coordinate rows.

19:40–21:10 / 18 Weights 24 and 32 Nonzero u coefficient: 3 times 8. Pure w coefficient: 2 times 16. Rank six.

21:10–22:10 / 19 Geometric family Recover 48 at $p = 4, q = 2$. Other counts are lower bounds. Star is a fixed-support census.

Column picture and completion *22:10–22:20: outline, current theme highlighted.*

22:20–23:40 / 20 Column indicators Occupied columns as a Boolean function. $[\![8,3,2]\!]$: y. Q48: $w_1 + w_2 + w_1 w_2$.

23:40–25:10 / 21 Moments and monomials Overlap parity is a monomial moment. One exceptional cubic. State the coset briefly.

25:10–26:40 / 22 Logical labels Three label bits per point. Product forces $x = \kappa$. S-only constraints already hold.

26:40–28:40 / 23 Logical completion One-row space, compatible pairs, linear third row. Three-site witness makes d_Z=3.

28:40–30:10 / 24 $T/T^{\dagger}$ signs Binary gamma in mod-4 equations. Show the polynomial, count $6 + 6 + 10$ inverse gates.

Bounds and questions *30:10–30:20: outline, current theme highlighted.*

30:20–32:10 / 25 Reed–Muller supports Positive-degree tests vanish at 0. Correct weight parity, then use RM duality.

32:10–33:40 / 26 Exclusion through 38 Rank at most 8, support reductions, complete enumeration, no logical completion.

33:40–35:40 / 27 Length comparison Our 38 floor. WJS 39–46. 47 quasi-transversal. Sign failure is code-specific.

35:40–37:10 / 28 Takeaways and questions Geometry, labels, signs. Ask about higher distance and sign feasibility. Leave QR up.

37:10: finish and invite questions.

Q&A / B1–B4 Backups Reed–Muller lengths, rank reduction, finite cases, dictionary with [CLM+26].